Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

Curious data scientist with 7 years' experience applying machine learning (ML) across healthcare, pharmaceutical analytics, manufacturing, e-commerce, and energy. I bring a background in ML and hands-on experience with statistical learning, probabilistic modelling, explainable AI, and production ML. I have a strong research interest in ML, mathematics, and their application in real-world algorithmic systems.

Education

2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. See thesis.
Relevant coursework: Statistical Machine Learning; Causality; Reinforcement Learning.

2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics**, *Exchange*

Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Built probabilistic, deep computer vision, dose-planning models from segmented 3D CT scans.
- Learned latent anatomical representations using autoencoders.
- Validated modelling choices with clinicians and medical physicists.

Keywords: deep learning, ML, probabilistic modelling, autoencoders, sparse Gaussian processes, optimisation, medical AI

Read more...

Full Stack Data Scientist @ Signum Life Science

- Architecting and building a Databricks forecasting platform for pharmaceutical analytics.
- Owned a next-best-action engine and extended explainable AI features.

Keywords: Databricks, explainable AI, forecasting

Read more...

Principal Data Scientist @ Valcon

- Built a production real-time neural network using natural-language inputs.
- Led explainable AI and Git trainings; mentored junior data scientists.
- Presented complex technical topics to non-technical stakeholders.

Keywords: deep learning, NLP, explainable AI, MLOps, stakeholder communication

Read more...

Skills and Other

Technology	Python, SQL, C++, PyTorch, SHAP, XGBoost, Databricks, AWS, MS Azure, FastAPI, MongoDB, Qdrant, Docker.
Language	Native English and Swedish; intermediate Danish.
Personal	In my spare time I enjoy working on projects, such as designing and building furniture.
Other	Valcon People's Choice Award (2023); Part of Team Sweden for IYPT 2015; Mensa Sweden.

Detailed Work Experience

2024–Present **Full-Stack Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, my current work is on an end-to-end forecasting platform for pharmaceutical price and demand forecasting, including patient population forecasts. The role spans architecture, project management, data science, ML engineering, and software engineering, with Databricks as a core platform.

I also maintain and improve a pharmaceutical price forecasting model for Denmark's fortnightly blind auctions. After stabilising the XGBoost-based system, I am now driving efforts toward a more robust and modern implementation.

Earlier in the role, I owned and extended Kwarts, a next-best-action prediction system for pharmaceutical sales representatives. Originally built externally, the system was extended with explainable AI features in collaboration with UX designers, and supported commercialisation by aiding sales representatives. Kwarts adapts to varied customer data maturity, from rule-based logic to deep learning, and is deployed in AWS for scalability.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was the lead data scientist and architect of the solution and responsible for implementing the end-to-end production pipeline.

I was awarded the "Valcon People's Choice Award" for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

Delivered ML solutions in e-commerce. Built time series forecasts using Facebook Prophet, and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs.

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using a combination of deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing a probabilistic framework for dose planning conditioned on patient geometry.

Alongside this, I redesigned some of RaySearch's lexicographic optimisation algorithms to better reflect clinical decision hierarchies, particularly in palliative care where trade-offs between tumour control and organ sparing are nuanced. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with both oncological standards and practical workflow requirements.

Other Detailed Experience

2025–2025 **Solo Project**, *Project Iris*

In this personal project I am building an end-to-end computer vision system for making product images on e-commerce sites interactive. The pipeline detects individual items using a YOLOv8 model, and embeds them semantically using OpenCLIP. The embeddings are then compared to support intelligent product linking across all images and products in a web store.

The system includes a full-stack setup with a FastAPI backend, MongoDB for persistence, a Qdrant index for vector search, and a custom-built Chromium plug-in that overlays clickable links directly onto web shop imagery in real time. The project also includes a dynamic scraping and indexing pipeline for ingesting and structuring product data from arbitrary e-commerce sites.

The project has been a learning opportunity for fine-tuning of deep learning models and practical ML engineering.